DBMS Mini Project

Software Dependency Risk & Vulnerability Impact Analyzer

Databases: MySQL and Neo4j (Graph NoSQL)

Submitted by

Anupam Tiwari (PES2UG24CS073)

Ayushman Dey (PES2UG24CS102)

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1. Problem Statement

Modern applications depend on software packages, and those packages depend on other packages. When a vulnerability is announced in one package, a flat list of who imports it is not enough. An application that never uses the package directly can still be exposed through a chain of dependencies, and often only some versions of a package are vulnerable.

An organisation with many applications cannot answer "which of our applications are affected, and why?" from a simple inventory table. This project stores applications, package versions and the dependency links between them, records known vulnerabilities, and calculates the blast radius of a vulnerability: every package version and application affected directly or indirectly, together with the version to upgrade to.

2. Brief

- **Users:** an Administrator (manages packages, applications and vulnerabilities) and a Security Analyst (runs impact analysis and views reports).
- **MySQL:** the authoritative store for all records. It enforces integrity through primary keys, foreign keys, CHECK and UNIQUE constraints, and uses transactions, triggers, views, indexes, a stored procedure and a stored function.
- **Neo4j:** holds the dependency graph (application, package version, vulnerability). "Who depends on X, directly or indirectly?" is a multi-hop question, which a graph traversal follows directly instead of joining a table to itself at every level. It also returns the full path, which the web interface can draw.
- **Link between the two:** the same ids (version_id, app_id, cve_id) are used in both databases.
- **Web application:** Next.js, with login and two roles, calling both databases.

3. ER Diagram

Figure 1 shows the ER model in Chen notation; the legend under the diagram explains the symbols, and the labels 1, N and M give the cardinality. Foreign keys are not drawn in an ER diagram; they appear when the model is converted into tables, as described below.

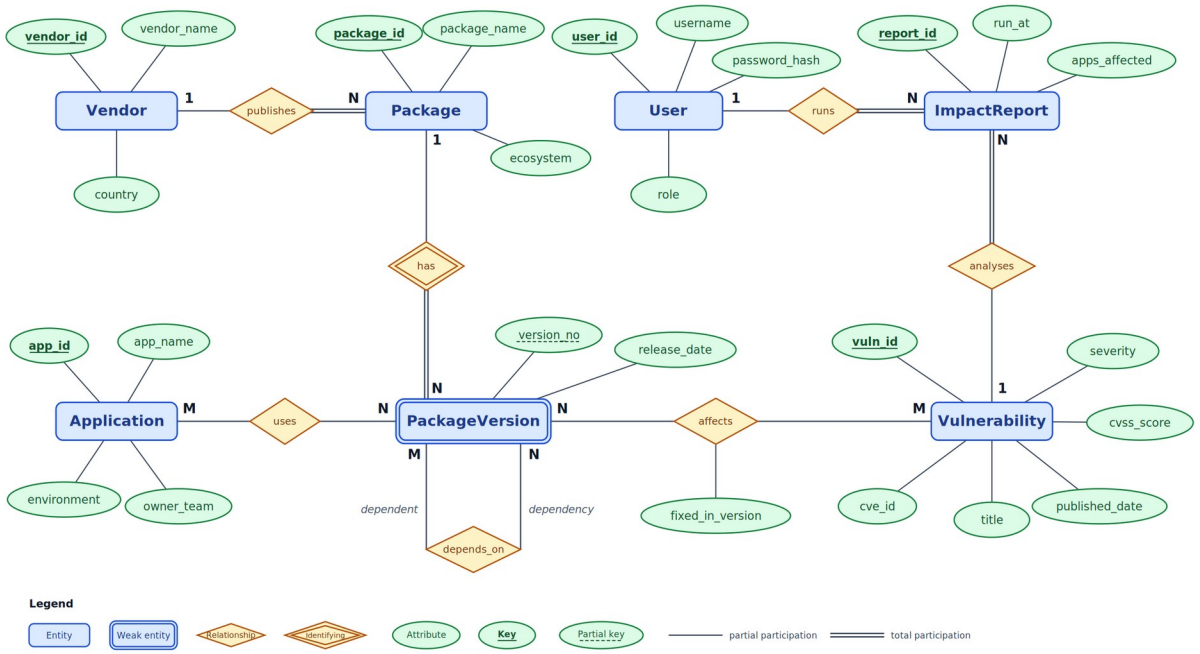


Figure 1: Overview of the ER model

Conversion to tables: every one-to-many relationship becomes a foreign key on the many side, for example packages.vendor_id. The foreign key is NOT NULL where participation is total (publishes, has, runs and analyses); the other relationships carry no such constraint. Each many-to-many relationship becomes a table: uses becomes app_dependencies, depends_on becomes package_dependencies and affects becomes vulnerability_affects. The only relationship attribute, fixed_in_version of affects, becomes the column fixed_in_version_id, a nullable foreign key to package_versions (NULL while no fixed version exists). The ER diagram names entities in the singular (Vendor, Package, ...); the MySQL tables are plural (vendors, packages, ...).

PackageVersion is the only weak entity. A version cannot exist without its package, and it is identified by the package together with its version number. In MySQL the table package_versions also has a surrogate key version_id, so that the relationships that refer to a version (uses, depends_on and affects) need only one column; the original identification is kept by UNIQUE (package_id, version_no), and ON DELETE CASCADE removes a package's versions together with it. All other entities have a key of their own and are strong: their foreign keys use NO ACTION.

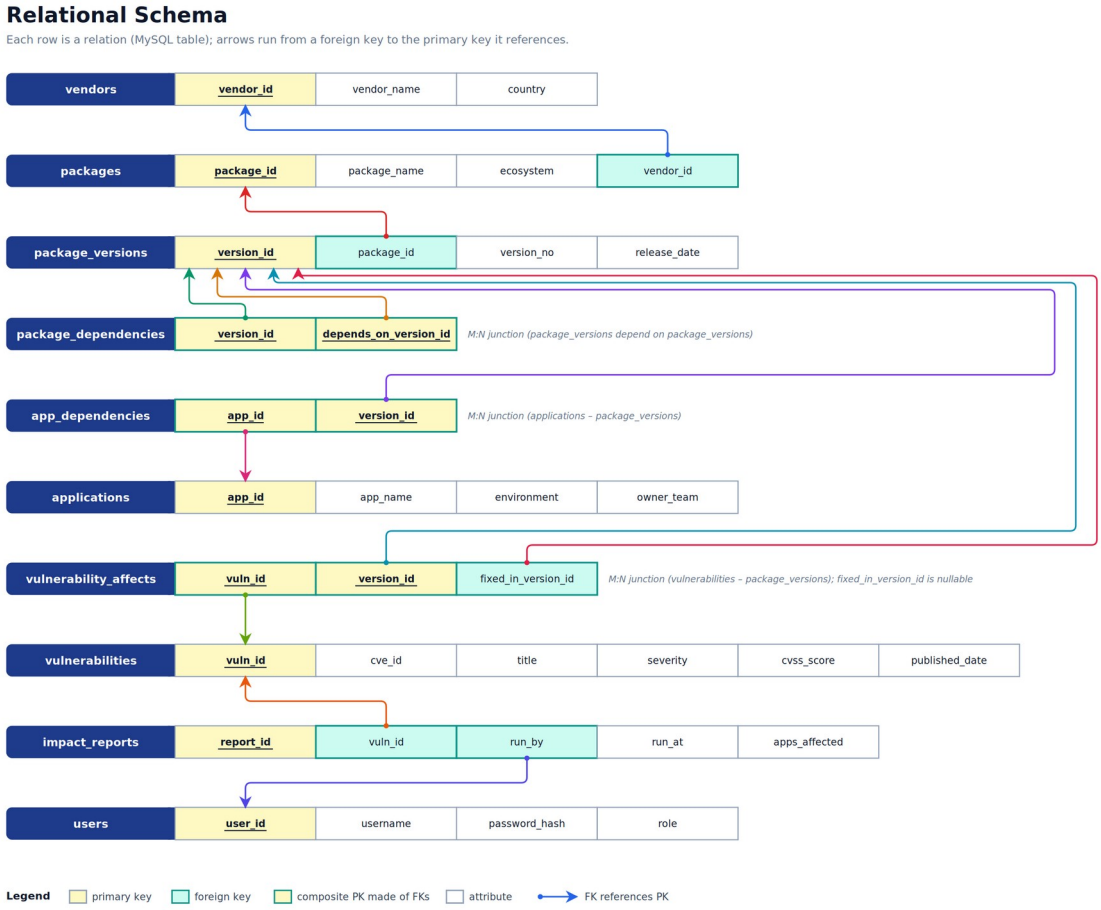
4. Relational Model and Relational Schema

Primary keys are underlined. Foreign keys are listed in the right column.

Relation	**Foreign keys**
users(user_id, username, password_hash, role)	-
vendors(vendor_id, vendor_name, country)	-
packages(package_id, package_name, ecosystem, vendor_id)	vendor_id → vendors
package_versions(version_id, package_id, version_no, release_date)	package_id → packages
applications(app_id, app_name, environment, owner_team)	-
app_dependencies(app_id, version_id)	app_id → applications version_id → package_versions
package_dependencies(version_id, depends_on_version_id)	version_id → package_versions depends_on_version_id → package_versions
vulnerabilities(vuln_id, cve_id, title, severity, cvss_score, published_date)	-
vulnerability_affects(vuln_id, version_id, fixed_in_version_id)	vuln_id → vulnerabilities version_id → package_versions fixed_in_version_id → package_versions
impact_reports(report_id, vuln_id, run_by, run_at, apps_affected)	vuln_id → vulnerabilities run_by → users

Other constraints: UNIQUE on username, vendor_name, app_name, cve_id, (package_name, ecosystem) and (package_id, version_no); CHECK cvss_score between 0 and 10; CHECK that a version cannot depend on itself. The columns role, ecosystem, environment and severity are restricted to fixed values with ENUM, and two BEFORE INSERT triggers reject two-way circular dependencies and a fixed version that belongs to a different package (Section 5.4).

Figure 2: Relational schema



5. DDL Commands

5.1 Database and tables

```
(PES2UG24CS102_PES2UG24CS073@root) [(none)]>CREATE DATABASE dependency_analyzer;
Query OK, 1 row affected (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [(none)]>USE dependency_analyzer;
Database changed
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TABLE users (
    ->     user_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     username VARCHAR(50) NOT NULL UNIQUE,
    ->     password_hash VARCHAR(255) NOT NULL,
    ->     role ENUM('ADMIN','ANALYST') NOT NULL DEFAULT 'ANALYST'
    -> );
Query OK, 0 rows affected (0.02 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TABLE vendors (
    ->     vendor_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     vendor_name VARCHAR(100) NOT NULL UNIQUE,
    ->     country VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.05 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TABLE packages (
    ->     package_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     package_name VARCHAR(100) NOT NULL,
    ->     ecosystem ENUM('npm','PyPI','Maven') NOT NULL,
    ->     vendor_id INT NOT NULL,
    ->     UNIQUE (package_name, ecosystem),
    ->     FOREIGN KEY (vendor_id) REFERENCES vendors(vendor_id)
    -> );
Query OK, 0 rows affected (0.02 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TABLE package_versions (
    ->     version_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     package_id INT NOT NULL,
    ->     version_no VARCHAR(20) NOT NULL,
    ->     release_date DATE NOT NULL,
    ->     UNIQUE (package_id, version_no),
    ->     FOREIGN KEY (package_id) REFERENCES packages(package_id)
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.02 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TABLE applications (
    ->     app_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     app_name VARCHAR(100) NOT NULL UNIQUE,
    ->     environment ENUM('PRODUCTION','INTERNAL','STAGING') NOT NULL,
    ->     owner_team VARCHAR(100) NOT NULL
    -> );
Query OK, 0 rows affected (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TABLE app_dependencies (
    ->     app_id INT NOT NULL,
    ->     version_id INT NOT NULL,
    ->     PRIMARY KEY (app_id, version_id),
    ->     FOREIGN KEY (app_id) REFERENCES applications(app_id)
    ->         ON DELETE CASCADE,
    ->     FOREIGN KEY (version_id) REFERENCES package_versions(version_id)
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TABLE package_dependencies (
    ->     version_id INT NOT NULL,
    ->     depends_on_version_id INT NOT NULL,
    ->     PRIMARY KEY (version_id, depends_on_version_id),
    ->     CHECK (version_id <> depends_on_version_id),
    ->     FOREIGN KEY (version_id) REFERENCES package_versions(version_id)
    ->         ON DELETE CASCADE,
    ->     FOREIGN KEY (depends_on_version_id) REFERENCES package_versions(version_id)
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TABLE vulnerabilities (
    ->     vuln_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     cve_id VARCHAR(20) NOT NULL UNIQUE,
    ->     title VARCHAR(200) NOT NULL,
    ->     severity ENUM('LOW','MEDIUM','HIGH','CRITICAL') NOT NULL,
    ->     cvss_score DECIMAL(3,1) NOT NULL CHECK (cvss_score BETWEEN 0 AND 10),
    ->     published_date DATE NOT NULL
    -> );
Query OK, 0 rows affected (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TABLE vulnerability_affects (
    ->     vuln_id INT NOT NULL,
    ->     version_id INT NOT NULL,
    ->     fixed_in_version_id INT NULL,
    ->     PRIMARY KEY (vuln_id, version_id),
    ->     FOREIGN KEY (vuln_id) REFERENCES vulnerabilities(vuln_id)
    ->         ON DELETE CASCADE,
    ->     FOREIGN KEY (version_id) REFERENCES package_versions(version_id)
    ->         ON DELETE CASCADE,
    ->     FOREIGN KEY (fixed_in_version_id) REFERENCES package_versions(version_id)
    ->         ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.02 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TABLE impact_reports (
    ->     report_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     vuln_id INT NOT NULL,
    ->     run_by INT NOT NULL,
    ->     run_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->     apps_affected INT NOT NULL,
    ->     FOREIGN KEY (vuln_id) REFERENCES vulnerabilities(vuln_id),
    ->     FOREIGN KEY (run_by) REFERENCES users(user_id)
    -> );
Query OK, 0 rows affected (0.02 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

5.2 Indexes

idx_pkgdep_reverse supports the "who depends on this version" lookup used at every step of the impact analysis.

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE INDEX idx_pkgdep_reverse
    -> ON package_dependencies (depends_on_version_id);
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE INDEX idx_appdep_version
    -> ON app_dependencies (version_id);
Query OK, 0 rows affected (0.00 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE INDEX idx_vulnaff_version
    -> ON vulnerability_affects (version_id);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE INDEX idx_vuln_severity
    -> ON vulnerabilities (severity);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

5.3 Views

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE VIEW v_vulnerable_versions AS
    -> SELECT
    ->     v.cve_id,
    ->     v.severity,
    ->     p.package_name,
    ->     pv.version_no,
    ->     CONCAT(fp.package_name, ' ', fv.version_no) AS fixed_in
    -> FROM vulnerabilities v
    -> JOIN vulnerability_affects va
    ->     ON va.vuln_id = v.vuln_id
    -> JOIN package_versions pv
    ->     ON pv.version_id = va.version_id
    -> JOIN packages p
    ->     ON p.package_id = pv.package_id
    -> LEFT JOIN package_versions fv
    ->     ON fv.version_id = va.fixed_in_version_id
    -> LEFT JOIN packages fp
    ->     ON fp.package_id = fv.package_id;
Query OK, 0 rows affected (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE VIEW v_app_footprint AS
    -> SELECT
    ->     a.app_name,
    ->     a.environment,
    ->     COUNT(ad.version_id) AS direct_dependencies
    -> FROM applications a
    -> LEFT JOIN app_dependencies ad
    ->     ON ad.app_id = a.app_id
    -> GROUP BY a.app_id, a.app_name, a.environment;
Query OK, 0 rows affected (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

5.4 Triggers

Two triggers. trg_pkgdep_before_insert rejects a two-way circular dependency between package versions. trg_vulnaff_before_insert rejects a vulnerability_affects row whose fixed version belongs to a different package than the affected version.

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DELIMITER //
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TRIGGER trg_pkgdep_before_insert
    -> BEFORE INSERT ON package_dependencies
    -> FOR EACH ROW
    -> BEGIN
    ->     IF EXISTS (
    ->         SELECT 1
    ->         FROM package_dependencies
    ->         WHERE version_id = NEW.depends_on_version_id
    ->           AND depends_on_version_id = NEW.version_id
    ->     ) THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Circular dependency detected';
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DELIMITER ;
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DELIMITER //
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TRIGGER trg_vulnaff_before_insert
    -> BEFORE INSERT ON vulnerability_affects
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.fixed_in_version_id IS NOT NULL
    ->        AND (SELECT package_id FROM package_versions
    ->             WHERE version_id = NEW.fixed_in_version_id)
    ->         <> (SELECT package_id FROM package_versions
    ->             WHERE version_id = NEW.version_id) THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Fixed version must belong to the same package';
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DELIMITER ;
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

5.5 Stored procedure and function

sp_impact_analysis takes a CVE id, computes the package versions affected directly or through any dependency chain using a recursive CTE, identifies the applications using those versions, and saves a row in impact_reports.

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DELIMITER //
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE PROCEDURE sp_impact_analysis(
    ->     IN p_cve VARCHAR(20),
    ->     IN p_user_id INT
    -> )
    -> BEGIN
    ->     DECLARE v_id INT;
    ->     DECLARE v_count INT;
    ->     SELECT vuln_id
    ->     INTO v_id
    ->     FROM vulnerabilities
    ->     WHERE cve_id = p_cve;
    ->     IF v_id IS NULL THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Unknown CVE id';
    ->     END IF;
    ->     DROP TEMPORARY TABLE IF EXISTS tmp_affected_apps;
    ->     CREATE TEMPORARY TABLE tmp_affected_apps (
    ->         app_id INT PRIMARY KEY
    ->     );
    ->     INSERT INTO tmp_affected_apps
    ->     WITH RECURSIVE affected (version_id) AS (
    ->         SELECT va.version_id
    ->         FROM vulnerability_affects va
    ->         WHERE va.vuln_id = v_id
    ->         UNION
    ->         SELECT pd.version_id
    ->         FROM package_dependencies pd
    ->         JOIN affected a
    ->             ON pd.depends_on_version_id = a.version_id
    ->     )
    ->     SELECT DISTINCT ad.app_id
    ->     FROM app_dependencies ad
    ->     WHERE ad.version_id IN (
    ->         SELECT version_id
    ->         FROM affected
    ->     );
    ->     SELECT COUNT(*)
    ->     INTO v_count
    ->     FROM tmp_affected_apps;
    ->     INSERT INTO impact_reports (vuln_id, run_by, apps_affected)
    ->     VALUES (v_id, p_user_id, v_count);
    ->     SELECT
    ->         a.app_name,
    ->         a.environment,
    ->         a.owner_team
    ->     FROM applications a
    ->     JOIN tmp_affected_apps t
    ->         ON t.app_id = a.app_id
    ->     ORDER BY FIELD(
    ->         a.environment,
    ->         'PRODUCTION',
    ->         'INTERNAL',
    ->         'STAGING'
    ->     ), a.app_name;
    -> END//
Query OK, 0 rows affected (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DELIMITER ;
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

fn_risk_level takes a CVSS score and returns the risk level as text: CRITICAL from 9.0, HIGH from 7.0, MEDIUM from 4.0 and LOW below that. It can be used in any query in place of a typed-out CASE expression.

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DELIMITER //
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE FUNCTION fn_risk_level(
    ->     p_cvss DECIMAL(3,1)
    -> )
    -> RETURNS VARCHAR(10)
    -> DETERMINISTIC
    -> BEGIN
    ->     IF p_cvss >= 9.0 THEN
    ->         RETURN 'CRITICAL';
    ->     ELSEIF p_cvss >= 7.0 THEN
    ->         RETURN 'HIGH';
    ->     ELSEIF p_cvss >= 4.0 THEN
    ->         RETURN 'MEDIUM';
    ->     ELSE
    ->         RETURN 'LOW';
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DELIMITER ;
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

5.6 Verification, ALTER and DROP

(1) Tables and views created

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SHOW FULL TABLES;
+------------------------------+-----------+
| Tables_in_dependency_analyzer | Table_type |
+------------------------------+-----------+
| app_dependencies              | BASE TABLE |
| applications                  | BASE TABLE |
| impact_reports                | BASE TABLE |
| package_dependencies          | BASE TABLE |
| package_versions              | BASE TABLE |
| packages                      | BASE TABLE |
| users                         | BASE TABLE |
| v_app_footprint               | VIEW       |
| v_vulnerable_versions         | VIEW       |
| vendors                       | BASE TABLE |
| vulnerabilities               | BASE TABLE |
| vulnerability_affects         | BASE TABLE |
+------------------------------+-----------+
12 rows in set (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

(2) Structure of users, vendors, packages, package_versions

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DESC users;
+---------------+------------------------+------+-----+---------+----------------+
| Field         | Type                   | Null | Key | Default | Extra          |
+---------------+------------------------+------+-----+---------+----------------+
| user_id       | int                    | NO   | PRI | NULL    | auto_increment |
| username      | varchar(50)            | NO   | UNI | NULL    |                |
| password_hash | varchar(255)           | NO   |     | NULL    |                |
| role          | enum('ADMIN','ANALYST') | NO   |     | ANALYST |                |
+---------------+------------------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DESC vendors;
+-------------+--------------+------+-----+---------+----------------+
| Field       | Type         | Null | Key | Default | Extra          |
+-------------+--------------+------+-----+---------+----------------+
| vendor_id   | int          | NO   | PRI | NULL    | auto_increment |
| vendor_name | varchar(100) | NO   | UNI | NULL    |                |
| country     | varchar(50)  | YES  |     | NULL    |                |
+-------------+--------------+------+-----+---------+----------------+
3 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DESC packages;
+--------------+---------------------------+------+-----+---------+----------------+
| Field        | Type                      | Null | Key | Default | Extra          |
+--------------+---------------------------+------+-----+---------+----------------+
| package_id   | int                       | NO   | PRI | NULL    | auto_increment |
| package_name | varchar(100)              | NO   | MUL | NULL    |                |
| ecosystem    | enum('npm','PyPI','Maven') | NO   |     | NULL    |                |
| vendor_id    | int                       | NO   | MUL | NULL    |                |
+--------------+---------------------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DESC package_versions;
+--------------+-------------+------+-----+---------+----------------+
| Field        | Type        | Null | Key | Default | Extra          |
+--------------+-------------+------+-----+---------+----------------+
| version_id   | int         | NO   | PRI | NULL    | auto_increment |
| package_id   | int         | NO   | MUL | NULL    |                |
| version_no   | varchar(20) | NO   |     | NULL    |                |
| release_date | date        | NO   |     | NULL    |                |
+--------------+-------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

(3) Structure of applications, app_dependencies, package_dependencies

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DESC applications;
+-------------+-----------------------------------------+------+-----+---------+----------------+
| Field       | Type                                    | Null | Key | Default | Extra          |
+-------------+-----------------------------------------+------+-----+---------+----------------+
| app_id      | int                                     | NO   | PRI | NULL    | auto_increment |
| app_name    | varchar(100)                            | NO   | UNI | NULL    |                |
| environment | enum('PRODUCTION','INTERNAL','STAGING') | NO   |     | NULL    |                |
| owner_team  | varchar(100)                            | NO   |     | NULL    |                |
+-------------+-----------------------------------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DESC app_dependencies;
+------------+------+------+-----+---------+-------+
| Field      | Type | Null | Key | Default | Extra |
+------------+------+------+-----+---------+-------+
| app_id     | int  | NO   | PRI | NULL    |       |
| version_id | int  | NO   | PRI | NULL    |       |
+------------+------+------+-----+---------+-------+
2 rows in set (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DESC package_dependencies;
+----------------------+------+------+-----+---------+-------+
| Field                | Type | Null | Key | Default | Extra |
+----------------------+------+------+-----+---------+-------+
| version_id           | int  | NO   | PRI | NULL    |       |
| depends_on_version_id | int  | NO   | PRI | NULL    |       |
+----------------------+------+------+-----+---------+-------+
2 rows in set (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

(4) Structure of vulnerabilities, vulnerability_affects, impact_reports

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DESC vulnerabilities;
+---------------+-----------------------------------+------+-----+---------+----------------+
| Field         | Type                              | Null | Key | Default | Extra          |
+---------------+-----------------------------------+------+-----+---------+----------------+
| vuln_id       | int                               | NO   | PRI | NULL    | auto_increment |
| cve_id        | varchar(20)                       | NO   | UNI | NULL    |                |
| title         | varchar(200)                      | NO   |     | NULL    |                |
| severity      | enum('LOW','MEDIUM','HIGH','CRITICAL') | NO   | MUL | NULL    |                |
| cvss_score    | decimal(3,1)                      | NO   |     | NULL    |                |
| published_date | date                              | NO   |     | NULL    |                |
+---------------+-----------------------------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DESC vulnerability_affects;
+--------------------+------+------+-----+---------+-------+
| Field              | Type | Null | Key | Default | Extra |
+--------------------+------+------+-----+---------+-------+
| vuln_id            | int  | NO   | PRI | NULL    |       |
| version_id         | int  | NO   | PRI | NULL    |       |
| fixed_in_version_id | int  | YES  | MUL | NULL    |       |
+--------------------+------+------+-----+---------+-------+
3 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DESC impact_reports;
+--------------+----------+------+-----+-------------------+-------------------+
| Field        | Type     | Null | Key | Default           | Extra             |
+--------------+----------+------+-----+-------------------+-------------------+
| report_id    | int      | NO   | PRI | NULL              | auto_increment    |
| vuln_id      | int      | NO   | MUL | NULL              |                   |
| run_by       | int      | NO   | MUL | NULL              |                   |
| run_at       | datetime | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| apps_affected | int      | NO   |     | NULL              |                   |
+--------------+----------+------+-----+-------------------+-------------------+
5 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

(5) Indexes, triggers, stored procedure and function created

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT index_name, table_name, column_name
    -> FROM information_schema.statistics
    -> WHERE table_schema = 'dependency_analyzer' AND index_name LIKE 'idx\_%';
+--------------------+----------------------+----------------------+
| INDEX_NAME         | TABLE_NAME           | COLUMN_NAME          |
+--------------------+----------------------+----------------------+
| idx_appdep_version | app_dependencies     | version_id           |
| idx_pkgdep_reverse | package_dependencies | depends_on_version_id |
| idx_vuln_severity  | vulnerabilities      | severity             |
| idx_vulnaff_version | vulnerability_affects | version_id          |
+--------------------+----------------------+----------------------+
4 rows in set (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT trigger_name, event_manipulation AS event,
    -> action_timing AS timing, event_object_table AS on_table
    -> FROM information_schema.triggers
    -> WHERE trigger_schema = 'dependency_analyzer';
+-------------------------+--------+--------+----------------------+
| TRIGGER_NAME            | event  | timing | on_table             |
+-------------------------+--------+--------+----------------------+
| trg_pkgdep_before_insert | INSERT | BEFORE | package_dependencies |
| trg_vulnaff_before_insert | INSERT | BEFORE | vulnerability_affects |
+-------------------------+--------+--------+----------------------+
2 rows in set (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT routine_name, routine_type
    -> FROM information_schema.routines
    -> WHERE routine_schema = 'dependency_analyzer';
+------------------+--------------+
| ROUTINE_NAME     | ROUTINE_TYPE |
+------------------+--------------+
| fn_risk_level    | FUNCTION     |
| sp_impact_analysis | PROCEDURE  |
+------------------+--------------+
2 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

(6) ALTER and DROP: add and remove a column, create and drop a table, drop and recreate an index

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>ALTER TABLE vendors
    -> ADD COLUMN website VARCHAR(100);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DESC vendors;
+-------------+--------------+------+-----+---------+----------------+
| Field       | Type         | Null | Key | Default | Extra          |
+-------------+--------------+------+-----+---------+----------------+
| vendor_id   | int          | NO   | PRI | NULL    | auto_increment |
| vendor_name | varchar(100) | NO   | UNI | NULL    |                |
| country     | varchar(50)  | YES  |     | NULL    |                |
| website     | varchar(100) | YES  |     | NULL    |                |
+-------------+--------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>ALTER TABLE vendors
    -> DROP COLUMN website;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE TABLE scratch_test (
    ->     id INT PRIMARY KEY
    -> );
Query OK, 0 rows affected (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DROP TABLE scratch_test;
Query OK, 0 rows affected (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DROP INDEX idx_vuln_severity
    -> ON vulnerabilities;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CREATE INDEX idx_vuln_severity
    -> ON vulnerabilities (severity);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SHOW TABLES LIKE 'scratch%';
Empty set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6. DML Commands

Each screenshot shows the commands and their output, run in the MySQL 8.0 client in the order given, on the database created in Section 5.

6.1 INSERT: users and vendors

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO users (username, password_hash, role) VALUES
    ->     ('admin1', '$2b$10$zrE5Uh.84sDAOvH2DJpC0uxsEhtjgd1NQ9ImdIlPdxlVJ1AFlHGyu', 'ADMIN'),
    ->     ('analyst1', '$2b$10$gnzUWV9EMaGXupPnmPXYd.mRH8wF.8Q16yrnAUjNvv33ldXDeggEa', 'ANALYST');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO vendors (vendor_name, country) VALUES
    ->     ('OpenForge', 'USA'),
    ->     ('NimbleSoft', 'India'),
    ->     ('CoreLabs', 'Germany');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT * FROM users;
+---------+----------+--------------------------------------------------------------+---------+
| user_id | username | password_hash                                                | role    |
+---------+----------+--------------------------------------------------------------+---------+
|       1 | admin1   | $2b$10$zrE5Uh.84sDAOvH2DJpC0uxsEhtjgd1NQ9ImdIlPdxlVJ1AFlHGyu | ADMIN   |
|       2 | analyst1 | $2b$10$gnzUWV9EMaGXupPnmPXYd.mRH8wF.8Q16yrnAUjNvv33ldXDeggEa | ANALYST |
+---------+----------+--------------------------------------------------------------+---------+
2 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT * FROM vendors;
+-----------+-------------+---------+
| vendor_id | vendor_name | country |
+-----------+-------------+---------+
|         1 | OpenForge   | USA     |
|         2 | NimbleSoft  | India   |
|         3 | CoreLabs    | Germany |
+-----------+-------------+---------+
3 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.2 INSERT: packages and package versions

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO packages (package_name, ecosystem, vendor_id) VALUES
    ->     ('logcore', 'Maven', 1),
    ->     ('jsonparse', 'npm', 1),
    ->     ('httpkit', 'npm', 2),
    ->     ('authlib', 'PyPI', 2),
    ->     ('webframe', 'npm', 2),
    ->     ('dbdriver', 'Maven', 3),
    ->     ('cryptolite', 'PyPI', 3),
    ->     ('uikit', 'npm', 1);
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO package_versions
    ->     (package_id, version_no, release_date)
    -> VALUES
    ->     (1, '1.0.0', '2024-01-10'),
    ->     (1, '1.2.0', '2026-03-02'),
    ->     (2, '2.0.0', '2024-05-20'),
    ->     (2, '2.1.0', '2026-04-11'),
    ->     (3, '3.4.0', '2024-09-01'),
    ->     (3, '3.5.0', '2026-04-20'),
    ->     (4, '1.0.0', '2024-11-15'),
    ->     (5, '4.0.0', '2024-10-05'),
    ->     (5, '4.1.0', '2026-05-01'),
    ->     (6, '5.2.0', '2024-08-08'),
    ->     (7, '0.9.0', '2023-12-12'),
    ->     (8, '1.0.0', '2025-02-14');
Query OK, 12 rows affected (0.00 sec)
Records: 12  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT
    ->     pv.version_id,
    ->     p.package_name,
    ->     pv.version_no,
    ->     pv.release_date
    -> FROM package_versions pv
    -> JOIN packages p
    ->     ON p.package_id = pv.package_id
    -> ORDER BY p.package_name, pv.version_no;
+------------+--------------+------------+--------------+
| version_id | package_name | version_no | release_date |
+------------+--------------+------------+--------------+
|          7 | authlib      | 1.0.0      | 2024-11-15   |
|         11 | cryptolite   | 0.9.0      | 2023-12-12   |
|         10 | dbdriver     | 5.2.0      | 2024-08-08   |
|          5 | httpkit      | 3.4.0      | 2024-09-01   |
|          6 | httpkit      | 3.5.0      | 2026-04-20   |
|          3 | jsonparse    | 2.0.0      | 2024-05-20   |
|          4 | jsonparse    | 2.1.0      | 2026-04-11   |
|          1 | logcore      | 1.0.0      | 2024-01-10   |
|          2 | logcore      | 1.2.0      | 2026-03-02   |
|         12 | uikit        | 1.0.0      | 2025-02-14   |
|          8 | webframe     | 4.0.0      | 2024-10-05   |
|          9 | webframe     | 4.1.0      | 2026-05-01   |
+------------+--------------+------------+--------------+
12 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.3 INSERT: applications and their direct dependencies

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO applications (app_name, environment, owner_team) VALUES
    ->     ('PaymentService', 'PRODUCTION', 'Payments'),
    ->     ('UserPortal', 'PRODUCTION', 'Web'),
    ->     ('AdminConsole', 'INTERNAL', 'Operations'),
    ->     ('ReportingTool', 'STAGING', 'Analytics'),
    ->     ('MobileAPI', 'PRODUCTION', 'Mobile');
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO app_dependencies (app_id, version_id) VALUES
    ->     (1, 8),
    ->     (1, 10),
    ->     (2, 9),
    ->     (2, 12),
    ->     (3, 7),
    ->     (4, 10),
    ->     (5, 7),
    ->     (5, 9);
Query OK, 8 rows affected (0.00 sec)
Records: 8  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT
    ->     a.app_name,
    ->     p.package_name,
    ->     pv.version_no
    -> FROM app_dependencies ad
    -> JOIN applications a
    ->     ON a.app_id = ad.app_id
    -> JOIN package_versions pv
    ->     ON pv.version_id = ad.version_id
    -> JOIN packages p
    ->     ON p.package_id = pv.package_id
    -> ORDER BY a.app_name, p.package_name;
+---------------+--------------+------------+
| app_name      | package_name | version_no |
+---------------+--------------+------------+
| AdminConsole  | authlib      | 1.0.0      |
| MobileAPI     | authlib      | 1.0.0      |
| MobileAPI     | webframe     | 4.1.0      |
| PaymentService | dbdriver    | 5.2.0      |
| PaymentService | webframe    | 4.0.0      |
| ReportingTool | dbdriver     | 5.2.0      |
| UserPortal    | uikit        | 1.0.0      |
| UserPortal    | webframe     | 4.1.0      |
+---------------+--------------+------------+
8 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

In app_dependencies each pair is (app_id, version_id).

6.4 INSERT: package-to-package dependencies

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO package_dependencies
    ->     (version_id, depends_on_version_id)
    -> VALUES
    ->     (5, 1),
    ->     (5, 3),
    ->     (6, 2),
    ->     (7, 5),
    ->     (7, 11),
    ->     (8, 5),
    ->     (9, 6),
    ->     (10, 1),
    ->     (12, 4);
Query OK, 9 rows affected (0.01 sec)
Records: 9  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT COUNT(*) AS dependency_edges
    -> FROM package_dependencies;
+------------------+
| dependency_edges |
+------------------+
|                9 |
+------------------+
1 row in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

Each pair is (version_id, depends_on_version_id). For example (5, 1) means httpkit 3.4.0 depends on logcore 1.0.0. Version ids are those listed in 6.2.

6.5 INSERT: vulnerabilities and affected versions

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO vulnerabilities
    ->     (cve_id, title, severity, cvss_score, published_date)
    -> VALUES
    ->     ('CVE-2026-1001', 'Remote code execution in logcore', 'CRITICAL', 9.8, '2026-06-15'),
    ->     ('CVE-2026-2002', 'Denial of service in jsonparse', 'HIGH', 7.5, '2026-07-02'),
    ->     ('CVE-2025-3003', 'Weak key derivation in cryptolite', 'MEDIUM', 5.3, '2025-11-20');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO vulnerability_affects
    ->     (vuln_id, version_id, fixed_in_version_id)
    -> VALUES
    ->     (1, 1, 2),
    ->     (2, 3, 4),
    ->     (3, 11, NULL);
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.6 TRANSACTION (COMMIT): register a new httpkit version with its dependencies

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO package_versions
    ->     (package_id, version_no, release_date)
    -> VALUES
    ->     (3, '3.6.0', '2026-08-15');
Query OK, 1 row affected (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO package_dependencies
    ->     (version_id, depends_on_version_id)
    -> VALUES
    ->     (LAST_INSERT_ID(), 2),
    ->     (LAST_INSERT_ID(), 4);
Query OK, 2 rows affected (0.00 sec)
Records: 2  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>COMMIT;
Query OK, 0 rows affected (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT
    ->     pv.version_id,
    ->     p.package_name,
    ->     pv.version_no,
    ->     pd.depends_on_version_id
    -> FROM package_versions pv
    -> JOIN packages p
    ->     ON p.package_id = pv.package_id
    -> JOIN package_dependencies pd
    ->     ON pd.version_id = pv.version_id
    -> WHERE pv.version_no = '3.6.0';
+------------+--------------+------------+----------------------+
| version_id | package_name | version_no | depends_on_version_id |
+------------+--------------+------------+----------------------+
|         13 | httpkit      | 3.6.0      |                    2 |
|         13 | httpkit      | 3.6.0      |                    4 |
+------------+--------------+------------+----------------------+
2 rows in set (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.7 TRANSACTION (ROLLBACK): a failed dependency insert undoes the whole registration

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>START TRANSACTION;
Query OK, 0 rows affected (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO package_versions
    ->     (package_id, version_no, release_date)
    -> VALUES
    ->     (6, '5.3.0', '2026-09-01');
Query OK, 1 row affected (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO package_dependencies
    ->     (version_id, depends_on_version_id)
    -> VALUES
    ->     (LAST_INSERT_ID(), 999);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`dependency_an
alyzer`.`package_dependencies`, CONSTRAINT `package_dependencies_ibfk_2` FOREIGN KEY (`depends_on_ve
rsion_id`) REFERENCES `package_versions` (`version_id`) ON DELETE CASCADE)
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT COUNT(*) AS dbdriver_530_rows
    -> FROM package_versions
    -> WHERE package_id = 6
    ->   AND version_no = '5.3.0';
+------------------+
| dbdriver_530_rows |
+------------------+
|                0 |
+------------------+
1 row in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

Version 999 does not exist, so the second insert fails. ROLLBACK then removes the version inserted before it, and the final count is 0.

6.8 TRIGGERS: two-way circular dependency and a fix in the wrong package are rejected

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO package_dependencies
    ->     (version_id, depends_on_version_id)
    -> VALUES
    ->     (2, 6);
ERROR 1644 (45000): Circular dependency detected
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

Version 6 (httpkit 3.5.0) already depends on version 2 (logcore 1.2.0), so the reverse link is rejected by the trigger.

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO vulnerability_affects
    ->     (vuln_id, version_id, fixed_in_version_id)
    -> VALUES
    ->     (1, 5, 2);
ERROR 1644 (45000): Fixed version must belong to the same package
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

Version 5 is httpkit 3.4.0 and version 2 is logcore 1.2.0, so the fixed version is in a different package and the second trigger rejects the row. The valid rows inserted in 6.5 pass the same trigger.

6.9 UPDATE: change the owner team of an application

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>UPDATE applications
    -> SET owner_team = 'Data Platform'
    -> WHERE app_name = 'ReportingTool';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT
    ->     app_name,
    ->     environment,
    ->     owner_team
    -> FROM applications;
+---------------+-------------+---------------+
| app_name      | environment | owner_team    |
+---------------+-------------+---------------+
| PaymentService | PRODUCTION  | Payments      |
| UserPortal     | PRODUCTION  | Web           |
| AdminConsole   | INTERNAL    | Operations    |
| ReportingTool  | STAGING     | Data Platform |
| MobileAPI      | PRODUCTION  | Mobile        |
+---------------+-------------+---------------+
5 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.10 DELETE: decommission an application (dependencies removed by ON DELETE CASCADE)

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO applications
    ->     (app_name, environment, owner_team)
    -> VALUES
    ->     ('LegacyBilling', 'INTERNAL', 'Finance');
Query OK, 1 row affected (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>INSERT INTO app_dependencies (app_id, version_id)
    -> SELECT app_id, 1
    -> FROM applications
    -> WHERE app_name = 'LegacyBilling';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT COUNT(*) AS rows_before
    -> FROM app_dependencies
    -> WHERE app_id = (
    ->     SELECT app_id
    ->     FROM applications
    ->     WHERE app_name = 'LegacyBilling'
    -> );
+-------------+
| rows_before |
+-------------+
|           1 |
+-------------+
1 row in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>DELETE FROM applications
    -> WHERE app_name = 'LegacyBilling';
Query OK, 1 row affected (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT COUNT(*) AS rows_after
    -> FROM app_dependencies
    -> WHERE app_id = 6;
+------------+
| rows_after |
+------------+
|          0 |
+------------+
1 row in set (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.11 SELECT (JOIN): remediation list - vulnerable versions and the version that fixes them

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT
    ->     v.cve_id,
    ->     v.severity,
    ->     p.package_name,
    ->     pv.version_no AS vulnerable_version,
    ->     CONCAT(fp.package_name, ' ', fv.version_no) AS upgrade_to
    -> FROM vulnerability_affects va
    -> JOIN vulnerabilities v
    ->     ON v.vuln_id = va.vuln_id
    -> JOIN package_versions pv
    ->     ON pv.version_id = va.version_id
    -> JOIN packages p
    ->     ON p.package_id = pv.package_id
    -> LEFT JOIN package_versions fv
    ->     ON fv.version_id = va.fixed_in_version_id
    -> LEFT JOIN packages fp
    ->     ON fp.package_id = fv.package_id
    -> ORDER BY v.cvss_score DESC;
+---------------+----------+-------------+--------------------+----------------+
| cve_id        | severity | package_name | vulnerable_version | upgrade_to     |
+---------------+----------+-------------+--------------------+----------------+
| CVE-2026-1001 | CRITICAL | logcore      | 1.0.0              | logcore 1.2.0  |
| CVE-2026-2002 | HIGH     | jsonparse    | 2.0.0              | jsonparse 2.1.0 |
| CVE-2025-3003 | MEDIUM   | cryptolite   | 0.9.0              | NULL           |
+---------------+----------+-------------+--------------------+----------------+
3 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.12 SELECT (GROUP BY, HAVING): most depended-on package versions

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT
    ->     p.package_name,
    ->     pv.version_no,
    ->     COUNT(*) AS direct_dependents
    -> FROM package_dependencies pd
    -> JOIN package_versions pv
    ->     ON pv.version_id = pd.depends_on_version_id
    -> JOIN packages p
    ->     ON p.package_id = pv.package_id
    -> GROUP BY
    ->     pv.version_id,
    ->     p.package_name,
    ->     pv.version_no
    -> HAVING COUNT(*) >= 2
    -> ORDER BY direct_dependents DESC;
+--------------+------------+-------------------+
| package_name | version_no | direct_dependents |
+--------------+------------+-------------------+
| httpkit      | 3.4.0      |                 2 |
| jsonparse    | 2.1.0      |                 2 |
| logcore      | 1.0.0      |                 2 |
| logcore      | 1.2.0      |                 2 |
+--------------+------------+-------------------+
4 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.13 SELECT (LEFT JOIN, aggregate): dependency footprint of each application

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT
    ->     a.app_name,
    ->     a.environment,
    ->     COUNT(ad.version_id) AS direct_dependencies
    -> FROM applications a
    -> LEFT JOIN app_dependencies ad
    ->     ON ad.app_id = a.app_id
    -> GROUP BY
    ->     a.app_id,
    ->     a.app_name,
    ->     a.environment
    -> ORDER BY direct_dependencies DESC, a.app_name;
+----------------+-------------+---------------------+
| app_name       | environment | direct_dependencies |
+----------------+-------------+---------------------+
| MobileAPI      | PRODUCTION  |                   2 |
| PaymentService | PRODUCTION  |                   2 |
| UserPortal     | PRODUCTION  |                   2 |
| AdminConsole   | INTERNAL    |                   1 |
| ReportingTool  | STAGING     |                   1 |
+----------------+-------------+---------------------+
5 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.14 SELECT (multi-table JOIN, aggregate): vendors ranked by number of vulnerable versions

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT
    ->     vd.vendor_name,
    ->     COUNT(DISTINCT pv.version_id) AS vulnerable_versions
    -> FROM vendors vd
    -> JOIN packages p
    ->     ON p.vendor_id = vd.vendor_id
    -> JOIN package_versions pv
    ->     ON pv.package_id = p.package_id
    -> JOIN vulnerability_affects va
    ->     ON va.version_id = pv.version_id
    -> GROUP BY
    ->     vd.vendor_id,
    ->     vd.vendor_name
    -> ORDER BY vulnerable_versions DESC;
+-------------+---------------------+
| vendor_name | vulnerable_versions |
+-------------+---------------------+
| OpenForge   |                   2 |
| CoreLabs    |                   1 |
+-------------+---------------------+
2 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.15 SELECT (subqueries): packages with no vulnerable version, and versions with above-average dependents

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT package_name
    -> FROM packages p
    -> WHERE NOT EXISTS (
    ->     SELECT 1
    ->     FROM package_versions pv
    ->     JOIN vulnerability_affects va
    ->         ON va.version_id = pv.version_id
    ->     WHERE pv.package_id = p.package_id
    -> );
+--------------+
| package_name |
+--------------+
| authlib      |
| dbdriver     |
| httpkit      |
| uikit        |
| webframe     |
+--------------+
5 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT
    ->     depends_on_version_id,
    ->     COUNT(*) AS dependents
    -> FROM package_dependencies
    -> GROUP BY depends_on_version_id
    -> HAVING COUNT(*) > (
    ->     SELECT AVG(c)
    ->     FROM (
    ->         SELECT COUNT(*) AS c
    ->         FROM package_dependencies
    ->         GROUP BY depends_on_version_id
    ->     ) t
    -> );
+----------------------+------------+
| depends_on_version_id | dependents |
+----------------------+------------+
|                    1 |          2 |
|                    2 |          2 |
|                    4 |          2 |
|                    5 |          2 |
+----------------------+------------+
4 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.16 SELECT (recursive CTE): every package version on a dependency chain from logcore 1.0.0

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>WITH RECURSIVE dependents AS (
    ->     SELECT
    ->         pv.version_id,
    ->         0 AS depth,
    ->         CAST(
    ->             CONCAT(p.package_name, ' ', pv.version_no)
    ->             AS CHAR(255)
    ->         ) AS path
    ->     FROM package_versions pv
    ->     JOIN packages p
    ->         ON p.package_id = pv.package_id
    ->     WHERE pv.version_id = 1
    ->     UNION ALL
    ->     SELECT
    ->         pd.version_id,
    ->         d.depth + 1,
    ->         CONCAT(
    ->             p2.package_name,
    ->             ' ',
    ->             pv2.version_no,
    ->             ' -> ',
    ->             d.path
    ->         )
    ->     FROM dependents d
    ->     JOIN package_dependencies pd
    ->         ON pd.depends_on_version_id = d.version_id
    ->     JOIN package_versions pv2
    ->         ON pv2.version_id = pd.version_id
    ->     JOIN packages p2
    ->         ON p2.package_id = pv2.package_id
    ->     WHERE d.depth < 10
    -> )
    -> SELECT depth, path
    -> FROM dependents
    -> ORDER BY depth, path;
+-------+-------------------------------------------------+
| depth | path                                            |
+-------+-------------------------------------------------+
|     0 | logcore 1.0.0                                   |
|     1 | dbdriver 5.2.0 -> logcore 1.0.0                 |
|     1 | httpkit 3.4.0 -> logcore 1.0.0                  |
|     2 | authlib 1.0.0 -> httpkit 3.4.0 -> logcore 1.0.0 |
|     2 | webframe 4.0.0 -> httpkit 3.4.0 -> logcore 1.0.0 |
+-------+-------------------------------------------------+
5 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

Each row shows a package version on a dependency chain from logcore 1.0.0, with the chain of dependencies that leads to it.

6.17 Views: querying v_vulnerable_versions and v_app_footprint

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT * FROM v_vulnerable_versions;
+---------------+----------+--------------+------------+-----------------+
| cve_id        | severity | package_name | version_no | fixed_in        |
+---------------+----------+--------------+------------+-----------------+
| CVE-2026-1001 | CRITICAL | logcore      | 1.0.0      | logcore 1.2.0   |
| CVE-2026-2002 | HIGH     | jsonparse    | 2.0.0      | jsonparse 2.1.0 |
| CVE-2025-3003 | MEDIUM   | cryptolite   | 0.9.0      | NULL            |
+---------------+----------+--------------+------------+-----------------+
3 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT * FROM v_app_footprint;
+----------------+-------------+---------------------+
| app_name       | environment | direct_dependencies |
+----------------+-------------+---------------------+
| PaymentService | PRODUCTION  |                   2 |
| UserPortal     | PRODUCTION  |                   2 |
| AdminConsole   | INTERNAL    |                   1 |
| ReportingTool  | STAGING     |                   1 |
| MobileAPI      | PRODUCTION  |                   2 |
+----------------+-------------+---------------------+
5 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.18 STORED PROCEDURE: blast radius of CVE-2026-1001

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CALL sp_impact_analysis('CVE-2026-1001', 1);
+----------------+-------------+---------------+
| app_name       | environment | owner_team    |
+----------------+-------------+---------------+
| MobileAPI      | PRODUCTION  | Mobile        |
| PaymentService | PRODUCTION  | Payments      |
| AdminConsole   | INTERNAL    | Operations    |
| ReportingTool  | STAGING     | Data Platform |
+----------------+-------------+---------------+
4 rows in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

UserPortal is not listed. It uses webframe 4.1.0, which depends on httpkit 3.5.0, and that in turn depends on logcore 1.2.0, the fixed version. Its other dependency, uikit 1.0.0, does not lead to the vulnerable version either.

6.19 STORED PROCEDURE: blast radius of CVE-2026-2002, then the saved reports

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>CALL sp_impact_analysis('CVE-2026-2002', 2);
+---------------+------------+-----------+
| app_name      | environment | owner_team |
+---------------+------------+-----------+
| MobileAPI     | PRODUCTION  | Mobile     |
| PaymentService | PRODUCTION  | Payments   |
| AdminConsole   | INTERNAL    | Operations |
+---------------+------------+-----------+
3 rows in set (0.02 sec)

Query OK, 0 rows affected (0.02 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT
    ->     r.report_id,
    ->     v.cve_id,
    ->     u.username AS run_by,
    ->     r.apps_affected
    -> FROM impact_reports r
    -> JOIN vulnerabilities v
    ->     ON v.vuln_id = r.vuln_id
    -> JOIN users u
    ->     ON u.user_id = r.run_by;
+-----------+--------------+----------+--------------+
| report_id | cve_id        | run_by   | apps_affected |
+-----------+--------------+----------+--------------+
|         1 | CVE-2026-1001 | admin1   |             4 |
|         2 | CVE-2026-2002 | analyst1 |             3 |
+-----------+--------------+----------+--------------+
2 rows in set (0.00 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

6.20 FUNCTION: risk level calculated from the CVSS score, next to the stored severity

```
(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>SELECT
    ->     cve_id,
    ->     cvss_score,
    ->     severity,
    ->     fn_risk_level(cvss_score) AS computed_level
    -> FROM vulnerabilities
    -> ORDER BY cvss_score DESC;
+---------------+------------+----------+----------------+
| cve_id        | cvss_score | severity | computed_level |
+---------------+------------+----------+----------------+
| CVE-2026-1001 |        9.8 | CRITICAL | CRITICAL       |
| CVE-2026-2002 |        7.5 | HIGH     | HIGH           |
| CVE-2025-3003 |        5.3 | MEDIUM   | MEDIUM         |
+---------------+------------+----------+----------------+
3 rows in set (0.01 sec)

(PES2UG24CS102_PES2UG24CS073@root) [dependency_analyzer]>
```

The function returns the same level as the severity stored for each CVE: 9.8 is CRITICAL, 7.5 is HIGH and 5.3 is MEDIUM.